Learning Mulligan Decisions in Magic: The Gathering via Probabilistic Modeling

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Abstract

Mulligan decisions in *Magic: The Gathering* represent a high-impact early-game choice under uncertainty, requiring players to evaluate the future playability of a hand based only on limited initial information. Despite their importance, these decisions are typically guided by informal heuristics rather than formal, data-driven models.

We model mulligan decisions as a probabilistic policy learning problem, estimating $\pi(S) = P(\text{Keep} \mid S)$ from observed player behavior. To support this, we construct a dataset of constructed-format mulligan decisions using a custom data collection pipeline and develop a structured feature representation capturing land count, card identity, mana curve, and play/draw context.

We evaluate logistic regression, XGBoost, and LightGBM models using 5-fold cross-validation under ROC-AUC and log loss. Across all models, we observe strong predictive performance (ROC-AUC > 0.90). Surprisingly, simple, interpretable features, particularly land count, capture the majority of the predictive signal, while more expressive nonlinear models provide only marginal gains. Ablation studies reveal that a low-dimensional mana-based representation explains most of the decision boundary, with card-level features acting primarily as refinements.

These results suggest that mulligan decisions are governed by a simple underlying structure dominated by resource consistency. More broadly, this work demonstrates that complex strategic decisions in combinatorial domains can often be approximated using compact, interpretable representations, providing a foundation for practical decision-support tools in competitive play.

1 Introduction

Mulligan decisions in *Magic: The Gathering* represent one of the earliest and most consequential choices in a game. Given an opening hand of seven cards, a player must decide whether to keep the hand or redraw at the cost of card disadvantage. This decision requires balancing immediate playability against long-term resource efficiency under uncertainty about future draws.

Despite its importance, the mulligan decision is typically guided by informal heuristics such as land-count rules or qualitative assessments of hand quality. This reliance on heuristics reflects the broader difficulty of modeling decision-making in Magic: The Gathering, a domain characterized by partial observability, stochastic transitions, and a combinatorial state space [9]. These heuristics are effective in practice but lack a formal representation, making it difficult to analyze, compare, or generalize decision-making across different decks and contexts.

From a decision-theoretic perspective, the optimal choice compares:

$$\mathbb{E}[\text{Win} \mid \text{keep}, S] \quad \text{vs.} \quad \mathbb{E}[\text{Win} \mid \text{mulligan}, S],$$

where S denotes the observable game state. However, estimating these expectations is intractable due to the combinatorial complexity of possible hands, hidden information, and the stochastic evolution of future game states. As a result, players rely on implicit approximations of this comparison.

In this work, we model mulligan decisions as a probabilistic policy learning problem. Rather than attempting to compute optimal decisions from first principles, we learn a policy

$$\pi(S) = P(\text{Keep} \mid S)$$

directly from observed player behavior. This data-driven formulation aligns with prior work using machine learning to analyze MTG decision-making and card-level data [7]. This formulation treats player decisions as noisy but informative signals of the underlying decision boundary.

A central challenge is representing card-based game states in a structured and learnable form. Opening hands exhibit high-dimensional combinatorial structure, with interactions between mana availability, card composition, and play/draw context. Additionally, the dataset is limited and may contain noise due to variation in player skill and preferences.

To address these challenges, we develop a feature representation that encodes early-game playability through land count, card identity, mana curve, and contextual features. We then evaluate both linear and nonlinear models to understand the structure of the decision boundary.

Our contributions are as follows:

- We introduce a data-driven formulation of mulligan decisions as a probabilistic policy learned from player behavior.
- We construct a dataset and feature representation that captures the combinatorial structure of opening hands while remaining interpretable.
- We provide a comparative evaluation of linear and nonlinear models, showing that simple models achieve competitive performance.
- We demonstrate through ablation studies that mulligan decisions are largely governed by low-dimensional mana-based features, with card-level information providing incremental refinements.

2 Related Work

Research on *Magic: The Gathering* (MTG) has explored several approaches to decision-making and representation learning, though none directly models mulligan decisions from player data.

Reinforcement learning and game-play agents. MTG has been studied as a challenging benchmark for reinforcement learning due to its partial observability, stochastic transitions, and extremely large action space. Recent work proposes RL frameworks combining self-play, search, and hierarchical action abstraction to learn game-play policies [9]. While these approaches aim to approximate optimal play, they require complex simulation environments and substantial computational resources. Moreover, they focus on full-game policies rather than isolated decision points, such as mulligans.

Strategy-based and heuristic agents. Alternative approaches rely on structured, domain-specific heuristics. For example, strategy-based agents evaluate opening hands using metrics such as "completable width" and "completable depth," which measure how well a hand supports predefined strategies [3]. While effective within specific domains, these methods depend heavily on handcrafted rules and do not learn directly from data, limiting their adaptability across decks and play styles.

Representation learning and drafting. A separate line of work focuses on learning representations of cards and decisions. Neural approaches have been used to learn generalizable card embeddings for drafting tasks, improving performance on unseen cards and formats [1]. However, these methods target deck construction decisions rather than in-game decision points, and do not address the dynamics of mulligan choices.

Interpretable and data-driven models. More broadly, there is growing interest in using interpretable models for decision-making tasks. Prior work argues that simple, transparent models can achieve competitive performance while offering greater insight into decision processes [8]. This perspective motivates the use of structured, low-dimensional representations in domains where interpretability is valuable.

Positioning of this work. In contrast to prior approaches, this work models mulligan decisions as a supervised learning problem using player-generated data. Rather than relying on simulation-based optimization (as in RL) or handcrafted heuristics, we directly learn the probability of keeping a hand from observed decisions. This isolates a single high-impact decision point and enables a focused analysis of its structure.

Our results suggest that, despite the combinatorial complexity of the underlying game, mulligan decisions are largely governed by a low-dimensional signal driven by mana consistency. This highlights a gap between the complexity of full-game modeling and the relative simplicity of specific decision components, providing a complementary perspective to existing MTG research.

3 Approach

3.1 Problem Formulation

We model the mulligan decision as a probabilistic policy over the game state. Let S denote the observable state at the time of the decision, consisting of the opening hand and contextual features such as play/draw.

We define the policy:

$$\pi(S) = P(\text{Keep} \mid S)$$

A decision is obtained via thresholding:

$$\text{Keep} \quad \text{if} \quad \pi(S) > \tau$$

From a decision-theoretic perspective, the optimal rule compares:

$$\mathbb{E}[\text{Win} \mid \text{Keep}, S] \quad \text{vs.} \quad \mathbb{E}[\text{Win} \mid \text{Mulligan}, S]$$

However, these expectations are not directly observable and are intractable to compute due to the combinatorial space of future game states and stochastic draws. Instead, we assume that observed player decisions provide noisy but informative labels reflecting this comparison.

Under this assumption, learning $\pi(S)$ corresponds to approximating the implicit decision boundary induced by player behavior.

3.2 Learning Objective

We treat mulligan decisions as a supervised binary classification problem. Given a dataset $\{(S_i, y_i)\}_{i=1}^n$, where $y_i \in \{0, 1\}$ indicates a keep decision, we learn a function $\pi(S)$ that minimizes the negative log-likelihood (log loss). This objective is standard in probabilistic classification and is widely used for estimating calibrated decision policies [5, 2].

This objective encourages accurate probabilistic predictions and penalizes overconfident errors. It is particularly appropriate for modeling decision policies, where calibrated probabilities are more informative than binary predictions.

For logistic regression, the policy takes the form:

$$\pi(S) = \sigma(w^\top x + b),$$

where x is the feature representation of S, and $\sigma(\cdot)$ is the logistic function. Tree-based models such as XGBoost and LightGBM are known to capture nonlinear interactions effectively in structured data [4, 6].

3.3 Data Collection Pipeline

A key challenge in this project is the absence of existing datasets for mulligan decisions in constructed *Magic: The Gathering*. Unlike draft or gameplay logs, mulligan decisions are not publicly recorded in a structured format. As a result, we developed a custom data collection tool to generate labeled training data.

The tool presents users with randomly sampled opening hands from a predefined decklist and prompts them to make a binary decision: *keep* or *mulligan*. Each decision is recorded along with the full hand composition and contextual features such as play/draw status.

This approach enables the collection of realistic, player-driven decision data while preserving the information available at the time of the mulligan decision. Importantly, the dataset reflects actual player heuristics rather than simulated or theoretically optimal play.

The data collection pipeline transforms a previously unobservable decision process into a supervised learning dataset, forming the foundation of the modeling approach.

Figure 1: Custom data collection interface used to generate mulligan decision data. Users are presented with randomly sampled opening hands and select whether to keep or mulligan, producing labeled training examples.

3.4 Feature Representation

We construct a structured representation of the opening hand designed to capture early-game playability.

Land Features. We include the total number of lands along with a quadratic term. This captures the nonlinear relationship between land count and hand quality, where both low and high land counts are undesirable.

Card Features. A bag-of-cards representation with step indicators encodes card identity and multiplicity. This allows the model to learn the marginal impact of specific cards and duplicated copies.

Mana Curve Features. Cards are grouped into mana value buckets, providing a coarse summary of the hand's curve and its ability to deploy threats efficiently.

Context Features. A binary play/draw indicator is included, as the value of a hand depends on whether the player goes first or second.

Together, these features map a high-dimensional combinatorial object (a hand of cards) into a structured, interpretable feature space.

3.5 Inductive Bias of Feature Design

The feature representation imposes a strong inductive bias: that mulligan decisions are primarily driven by early-game resource consistency.

Land count acts as a proxy for mana availability, determining whether a hand can execute early plays. The inclusion of a quadratic term encodes the hypothesis that both insufficient and excessive resources reduce hand quality.

Card-level features model deviations from this baseline, capturing how specific cards or combinations modify the value of an otherwise playable hand.

This induces a decomposition of the decision function:

$$\text{Decision Quality} \approx f(\text{mana consistency}) + g(\text{card composition})$$

This structure assumes that mana consistency provides the dominant signal, while card-level effects act as refinements. As shown in Section 4, ablation studies empirically validate this assumption.

4 Experiments

4.1 Model Performance

All models achieve strong predictive performance under 5-fold cross-validation, with ROC-AUC values exceeding 0.90. Tree-based models (XGBoost and LightGBM) slightly outperform logistic regression, suggesting the presence of limited nonlinear interactions. This is consistent with prior findings that gradient boosting methods can capture higher-order interactions in tabular datasets [4, 6].

However, the performance gap is small. Logistic regression achieves comparable results despite its linear structure, indicating that most predictive signals are captured by the engineered features rather than complex model architectures. This supports the perspective that interpretable models can achieve competitive performance in structured decision problems [8].

The baseline model (always predicting "keep") performs significantly worse, confirming that the models are learning meaningful structure rather than exploiting class imbalance.

Model	ROC-AUC	Log Loss	Accuracy
Always Keep (Baseline)	0.500	0.660	0.627
Logistic Regression	0.928	**0.320**	**0.848**
XGBoost	0.922	0.355	0.841
LightGBM	**0.931**	0.335	0.841

Table 1: Cross-validated performance comparison on the full feature set. The baseline model (always predicting "keep") achieves ROC-AUC of 0.5, confirming no discriminative ability. All learned models significantly outperform this baseline.

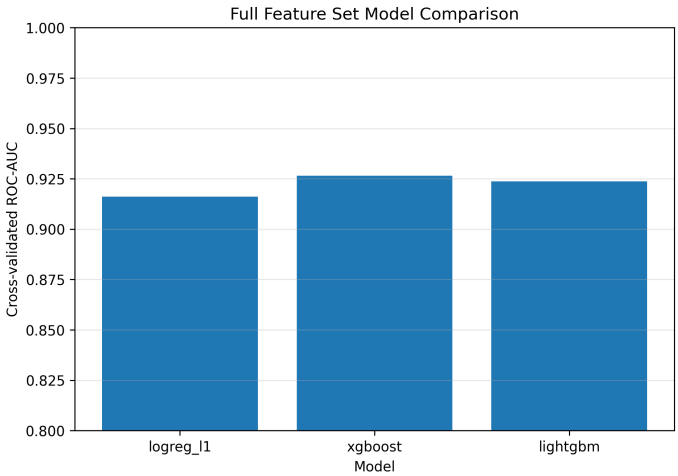

Figure 2: Model comparison on the full feature set. All models achieve similar performance, with only marginal improvements from nonlinear methods.

4.2 Ablation Study

To understand the contribution of different feature groups, we perform ablation experiments across all models.

Removing mana bucket features results in only a minor degradation in performance (e.g., $\Delta\text{AUC} \approx -0.005$ for logistic regression), indicating that these features provide limited additional signal beyond other representations.

In contrast, restricting the model to land-based features leads to a more substantial drop (ΔAUC ≈ -0.05), while a cards-only representation causes severe degradation (ΔAUC ≈ -0.30). This demonstrates that a large portion of the decision boundary can be explained by a low-dimensional representation dominated by mana availability.

Notably, this pattern is consistent across all model classes, suggesting that the observed structure is intrinsic to the decision problem rather than an artifact of a specific model.

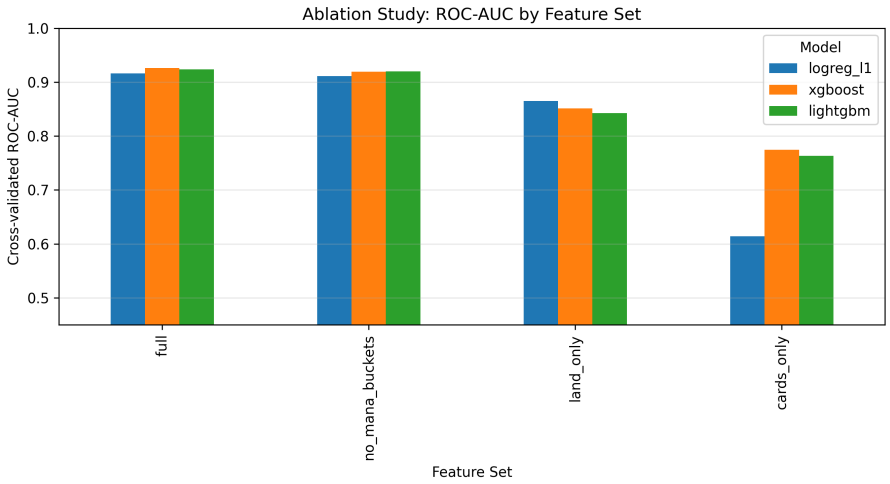

Figure 3: Ablation study showing ROC-AUC across feature sets. Performance remains high when removing mana bucket features but drops significantly for land-only and cards-only models, indicating that land-based features capture the dominant signal.

4.3 Structural Insights

The ablation results reveal a clear structural property of the decision problem: mulligan decisions are largely governed by a low-dimensional signal driven by mana availability.

The strong performance of the land-only model indicates that players rely heavily on resource consistency when evaluating opening hands. This aligns with domain knowledge, where hands with insufficient or excessive lands are typically unplayable.

The poor performance of the cards-only model further highlights that card quality alone is not sufficient to determine hand viability. Without sufficient mana, even high-impact cards cannot be effectively utilized.

The relatively small improvement from tree-based models suggests that higher-order interactions exist but are limited in magnitude. Most of the predictive structure is captured by simple, interpretable features. This further reinforces the argument for using interpretable models in high-stakes decision settings [8].

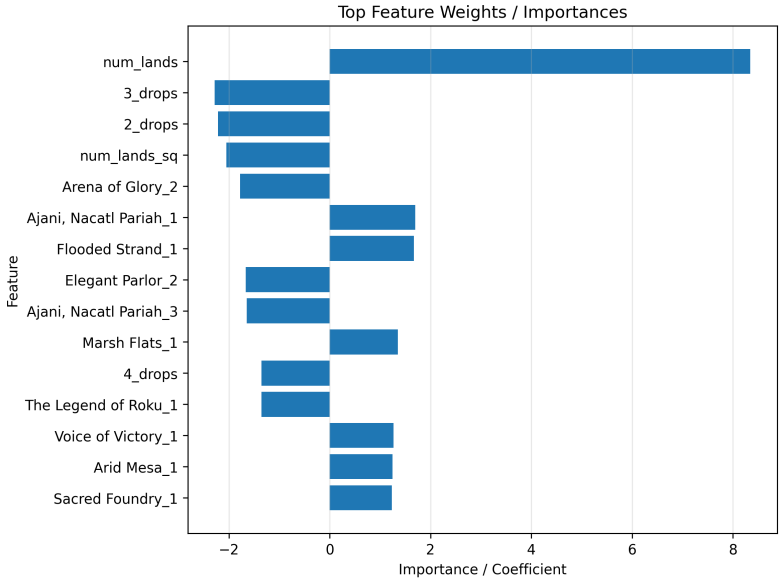

Figure 4: Top feature weights from logistic regression. Land count dominates the decision boundary, with additional refinements from mana curve and card-specific features.

4.4 Interpretation

The model learns a nonlinear relationship between land count and decision outcome, with 2–3 lands maximizing the probability of keeping a hand. Both low and high land counts are penalized, reflecting the importance of mana consistency.

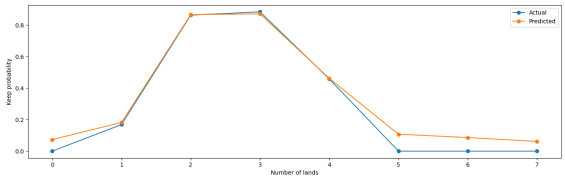

Figure 5: Actual and predicted keep probability as a function of land count. The model captures a nonlinear relationship with peak probability at 2–3 lands, closely matching empirical player behavior.

5 Discussion

5.1 Decomposing Mulligan Decisions

The empirical results suggest that mulligan decisions can be decomposed into a dominant land-based component and a secondary card-based adjustment.

The land-only model achieves ROC-AUC above 0.85, while the full model exceeds 0.92, indicating that a large portion of the decision boundary can be explained by mana availability alone. In contrast, the cards-only model performs substantially worse (ROC-AUC ≈ 0.6), demonstrating that card-level features require a strong mana baseline to be meaningful.

This supports a decomposition of the form:

$$\pi(S) = \pi_{\text{land}}(S) + \delta_{\text{card}}(S),$$

where π_{land} captures the primary decision signal and δ_{card} represents refinements based on card interactions. Empirically, π_{land} accounts for the majority of predictive performance, while δ_{card} provides marginal improvements in borderline cases.

5.2 Key Insights

The primary finding is that mulligan decisions are dominated by mana consistency. This aligns with prior work highlighting the importance of resource constraints and early-game structure in MTG decision-making [3]. The model learns a strongly nonlinear relationship between land count and decision outcome, with hands containing 2–3 lands having the highest probability of being kept.

Card-level features provide additional predictive power, but primarily refine decisions already determined by mana. This suggests a two-stage decision process: players first evaluate whether a hand is playable based on resource availability, and then consider the quality and synergy of specific cards.

The small performance gap between logistic regression and tree-based models further indicates that higher-order interactions are limited in magnitude. Most of the predictive structure is captured by low-dimensional, interpretable features, suggesting that the underlying decision boundary is relatively simple despite the high-dimensional input space.

5.3 Limitations

This study has several limitations.

First, the dataset is relatively small and may not fully capture the diversity of player behavior. The model is trained on player-labeled decisions, which introduces noise due to differences in skill level, play style, and strategic preferences.

Second, the model does not explicitly condition on deck archetype or matchup context. Mulligan decisions are often highly dependent on deck strategy (e.g., aggressive vs. control decks), and collapsing across archetypes may obscure important structure.

Third, the formulation assumes that observed decisions approximate optimal play. In practice, player decisions may be suboptimal, meaning that the learned policy reflects behavior rather than true optimality.

Finally, the feature representation focuses on static hand properties and does not capture deeper strategic considerations such as long-term game plans, sequencing flexibility, or hidden information.

5.4 Future Work

Several directions could extend this work.

A natural next step is to incorporate deck-level structure by conditioning on archetype or learning embeddings of deck composition. This would allow the model to distinguish between different strategic contexts and improve generalization.

Another direction is to model interactions between cards more explicitly, for example using graph-based or neural representations that capture synergy and combinations beyond simple bag-of-cards features.

From a decision-theoretic perspective, an important extension is to move beyond behavior modeling and estimate expected win probability directly. This would enable learning policies that approximate optimal play rather than observed decisions.

Finally, scaling data collection across a larger player base would allow for more robust estimation, improved generalization, and analysis of how mulligan strategies vary across skill levels.

5.5 Conclusion

This work demonstrates that mulligan decisions can be effectively modeled using simple, interpretable features. Despite the combinatorial complexity of the input space, the dominant signal is low-dimensional and driven primarily by mana consistency.

The results show that most predictive power can be captured by land-based features, while card-level features provide secondary refinements. This suggests that complex strategic decisions in *Magic: The Gathering* can often be decomposed into a small number of dominant structural factors.

These findings provide a foundation for future decision-support tools and highlight the potential for combining domain knowledge with data-driven modeling to better understand strategic decision-making in complex games.

Code Availability

All code used in this project is publicly available to support reproducibility and further research.

- Mulligan data collection tool: https://github.com/londonglenn/mtg_mulligan_tool
- Model training pipeline: https://github.com/londonglenn/mtg-mulligan-model
- Project blog (interactive demo and visualizations): https://londonglenn.github.io/mtg-mulligan-blog/

References

[1] Timo Bertram, Johannes Fürnkranz, and Martin Müller. Learning with generalised card representations for magic: The gathering. *arXiv*, 2024.

[2] Christopher M. Bishop. *Pattern Recognition and Machine Learning*. Springer, 2006.

[3] Michael Bowling, Shaelyn Rivers-Green, Deric Siglin, Lori Alvin, and Chris Alvin. Toward a competitive agent framework for magic: The gathering. *The Florida AI Research Society*, 2021.

[4] Tianqi Chen and Carlos Guestrin. Xgboost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 785–794, 2016.

[5] Trevor Hastie, Robert Tibshirani, and Jerome Friedman. *The Elements of Statistical Learning*. Springer, 2009.

[6] Guolin Ke, Qi Meng, Thomas Finley, Taifeng Wang, Wei Chen, Weidong Ma, Qiwei Ye, and Tie-Yan Liu. Lightgbm: A highly efficient gradient boosting decision tree. In *Advances in Neural Information Processing Systems*, 2017.

[7] Todd W. Neller. Ai education matters: Data science and machine learning with magic: The gathering. *AI Matters*, 5(2):8–10, 2019.

[8] Cynthia Rudin. Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead. *Nature Machine Intelligence*, 1(5):206–215, 2019.

[9] Alex Thaikanov, Blair Probst, Casey Tzao, and Dana Evelyn. Reinforcement learning for competitive magic: The gathering gameplay. *arXiv*, 2024.